

2. Probability

* Sample Space: A set of all possible outcomes of an experiment is called a Sample Space (A well defined objects is known as set)

Example:

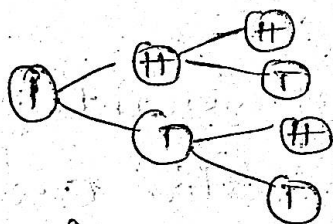
→ In throwing a die there are 6 possibilities

$$S = \{1, 2, 3, 4, 5, 6\}$$

→ A coin is tossed one time

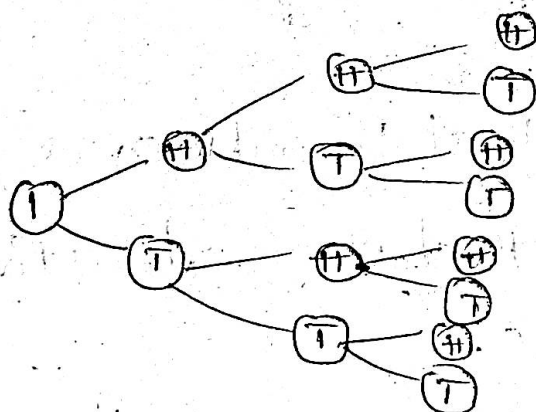
$$S = \{H, T\}$$

→ A coin is tossed twice



$$S = \{H, H, H, T, T, H, T, T\}$$

→ A coin is tossed three times



* Mutually Exclusive Events:

It two or more of them cannot happen simultaneously in the same trial.

Example: In throwing a die all possible 6 cases are mutually exclusive

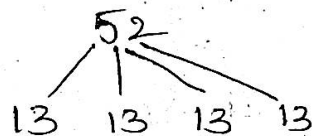
* Exhaustive

* Exhaustive Events:

The total number of possible outcomes in any trial is known as Exhaustive Events.

For Example: In tossing a coin there are two exhaustive cases i.e., Head or tail.

→ In drawing two cards from pack of cards the exhaustive number of cases is $52C_2$



* Probability: If an experiment is performed 'n' is the number of exhaustive cases and 'm' is the number of favorable cases of an event E. Then the probability of an event E is defined by

$$P(E) = \frac{\text{No. of Favorable cases}}{\text{Total no. of exhaustive cases}} = \frac{m}{n}$$

$$P = \frac{n(A)}{n(S)}$$

* Axioms of probability:

→ $0 \leq P(A) \leq 1$ for each event $A \subseteq S$

→ $P(S) = 1$

→ If A and B are any two mutually exclusive events then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{Or} \\ = P(A) + P(B) \quad [\because P(A \cap B) = 0]$$

* Conditional Probability:

If A and B are two events then the probability of the happening of the event B given that A has already happened is denoted by $P(B|A)$ and is defined as.

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \text{if } P(A) \neq 0$$

Similarly

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{if } P(B) \neq 0$$

Notes:

→ If A and B are events then $P(A \cup B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

→ If A, B and C are the events then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

* Independent Events:

If A and B are Independent then

$$P(A \cap B) = P(A) \cdot P(B)$$

* Addition law of probability:

For n events $A_1, A_2, \dots, A_n$ then $P(\bigcup_{i=1}^n A_i)$

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) + \sum_{1 \leq i < j < k \leq n} P(A_i \cap A_j \cap A_k) - \dots + (-1)^{n+1} P(A_1 \cap A_2 \cap \dots \cap A_n)$$

$$P(A_i \cap A_j \cap A_k) + \dots + (-1)^{n+1} P(A_1 \cap A_2 \cap \dots \cap A_n)$$

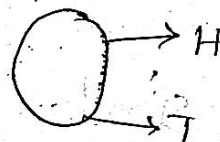
* Multiplicative law of probability:

$$P\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i)$$

1. Find the probability that exactly one head appears in a single throw of a fair coin

Let $S = \{H, T\}$

$$n(S) = 2$$



$A = \text{getting one head} = 1$

$$n(A) = 1$$

$$\text{Required probability } P(A) = \frac{n(A)}{n(S)} = \frac{1}{2}$$

2. Two dice are rolled at what is the probability that the sum of the upturned faces will be equal to 7?

$$6^n = 6^2 = 36$$

$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6),$
 $(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6),$
 $(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6),$
 $(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6),$
 $(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6),$
 $(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$

$$n(S) = 36 ; n(A) = \text{upturned faces} = 6$$

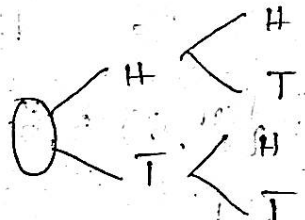
We know that

$$p(A) = \frac{n(A)}{n(S)} = \frac{6}{36} = \underline{\underline{\frac{1}{6}}}$$

3. If two coins are tossed what is the probability of getting one head.

$$S = \{HH, HT, TH, TT\}$$

$$n(S) = 4$$



$$A = \text{getting one head} = HT, TH$$

$$n(A) = 2$$

We know that

$$p(A) = \frac{n(A)}{n(S)} = \frac{2}{4} = \underline{\underline{\frac{1}{2}}}$$

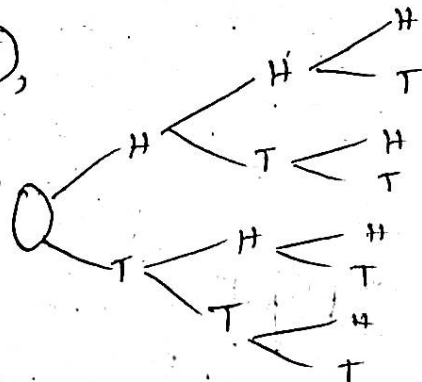
$$p(\text{getting one tail}) = \underline{\underline{\frac{1}{2}}}$$

$$p(\text{getting atleast one head and one tail}) = \underline{\underline{\frac{3}{4}}}$$

4. If 3 coins are tossed find the probability of getting $\rightarrow$ 3 heads, 2 heads, no heads

$$S = \{HHH, HHT, HTH, HTT, \\ THH, THT, TTH, TTT\}$$

$$n(S) = 8$$



$$\rightarrow p(\text{getting 3 heads}) = \frac{n(A)}{n(S)} = \frac{1}{8}$$

$$\rightarrow P(\text{getting 2 heads}) = \frac{n(A)}{n(S)} = \frac{3}{8}$$

$$\rightarrow P(\text{getting No heads}) = \frac{n(A)}{n(S)} = \frac{1}{8}$$

5) Find the probability of getting sum of two (10) if we throw two dice. Find the

$$\frac{3}{36} = \frac{1}{12}$$

$$\{(4,6), (5,5), (6,4)\}$$

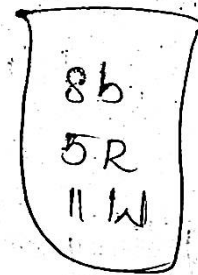
6. A bag contains 5 red balls, 8 blue balls and 11 white balls. 3 balls are gone together from the box. Find the probability that

→ one red, one blue, one white

→ two white and one red

→ three white

$$n(S) = 24$$



3 balls are gone together from the bag = ${}^{24}C_3$

$$\rightarrow P(\text{getting 1 red, 1 blue, 1 white}) = \frac{{}^5C_1 \times {}^8C_1 \times {}^{11}C_1}{{}^{24}C_3}$$

$$= \frac{5 \times 8 \times 11}{24 \times 23 \times 22}$$

$$= \frac{440}{2024}$$

$$= \frac{440}{2024} = 0.2174$$

→ P(getting two white, 1 red)

$$= \frac{{}^{11}C_2 \times {}^5C_1}{{}^{24}C_3} = 0.13586$$

$$\rightarrow P(\text{getting 3 white balls}) = \frac{{}^{11}C_3}{{}^{24}C_3} = \underline{0.4891}$$

1. A class consist of 6 girls and 10 boys if a Committee of 3 is choosen at random from the class find the probability that

$\rightarrow$ 3 boys are seluted (3, 14)

$\rightarrow$ Exactly 2 girls (15/16)

$$n(S) = 6 \text{ girls} + 10 \text{ boys} = 16$$

3 is choosen at random from the class = ${}^{16}C_3$

$$\rightarrow P(\text{getting 3 boys}) = \frac{{}^{10}C_3}{{}^{16}C_3} = \frac{\frac{10 \times 9 \times 8}{3 \times 2 \times 1}}{\frac{16 \times 15 \times 14}{3 \times 2 \times 1}} = \frac{720}{3360}$$

Baye's Theorem:

Statement:

Suppose $E_1, E_2, \dots, E_n$ are mutually exclusive events of a Sample Space such that

$P(E_i) > 0, i = 1, 2, \dots, n$ and A is any arbitrary event of S such that $P(A) > 0$ and $A \subset \bigcup_{i=1}^n E_i$

Then the conditional probability of E_i given $P(A) > 0, i = 1, 2, \dots, n$

equal to

$$P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum_{i=1}^n P(E_i)P(A|E_i)}$$

Proof:

Let S be the sample space of the random experiment the events $E_1, E_2, \dots, E_n$ being exhaustive events

$$S = E_1 \cup E_2 \cup \dots \cup E_n$$

$$= \bigcup_{i=1}^n E_i$$

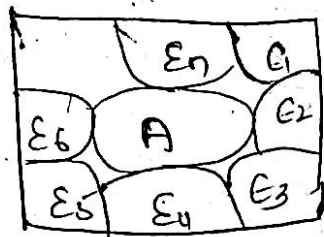
$$\text{Given } A \subset \bigcup_{i=1}^n E_i$$

$$A \subset S$$

We know that $A = A \cap S$ [$\because A \subset B \Rightarrow A \cap B = A$]

$$A = A \cap \left[\bigcup_{i=1}^n E_i \right]$$

$$= A \cap [E_1 \cup E_2 \cup \dots \cup E_n]$$



$$A = (A \cap E_1) \cup (A \cap E_2) \cup \dots \cup (A \cap E_n)$$

$$A = \bigcup_{i=1}^n (A \cap E_i)$$

$$P(A) = P\left[\bigcup_{i=1}^n (A \cap E_i)\right]$$

$$P(A) = \sum_{i=1}^n P(A \cap E_i)$$

Since $A \cap E_i, i=1, 2, \dots, n$ are mutually exclusive

$$P(A \cap E_i) = P(A|E_i) P(E_i) \rightarrow \text{By multiplication}$$

$$P(A) = \sum_{i=1}^n P(A|E_i) P(E_i)$$

By conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Now

$$P(E_i|A) = \frac{P(A \cap E_i)}{P(A)}$$

$$= \frac{P(E_i) P(A|E_i)}{\sum_{i=1}^n P(A|E_i) P(E_i)}$$

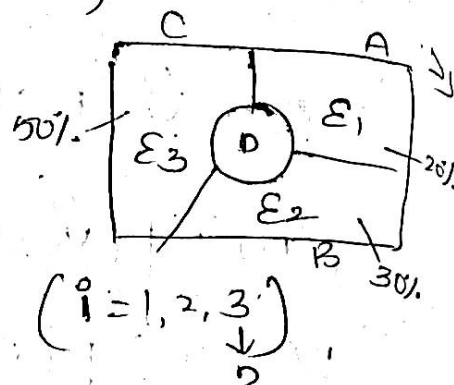
$$\sum_{i=1}^n P(A|E_i) P(E_i)$$

- In a bolt factory machines A, B, C manufactured 20%, 30%, 50% of the total of their output and 6%, 3%, and 2% are defective. A bolt is drawn at Random found to be defective.

What is the probability that i.e., manufactured by machines A, B, C

Sol We know that, Baye's Theorem

$$P(E_i|A) = \frac{P(E_i) P(A|E_i)}{\sum_{i=1}^n P(E_i) P(A|E_i)}$$



$$P(E_i|D) = \frac{P(E_i) P(D|E_i)}{\sum_{i=1}^3 P(E_i) P(D|E_i)} \quad (i=1, 2, 3)$$

$$= \frac{P(E_i) P(D|E_i)}{P(E_1) P(D|E_1) + P(E_2) P(D|E_2) + P(E_3) P(D|E_3)}$$

$$= \frac{P(E_i) P(D|E_i)}{P(A)}$$

$$P(E_1) = P(A) = 20\% = \frac{20}{100} = 0.2$$

$$P(E_2) = P(B) = 30\% = \frac{30}{100} = 0.3$$

$$P(E_3) = P(C) = 50\% = \frac{50}{100} = 0.5$$

$$P(D|E_1) = P(D|A) = 6\% = \frac{6}{100} = 0.06$$

$$P(D|E_2) = P(D|B) = 3\% = \frac{3}{100} = 0.03$$

$$P(D|E_3) = P(D|C) = 2\% = \frac{2}{100} = 0.02$$

$$P(A) = (0.2)(0.06) + (0.3)(0.03) + (0.5)(0.02)$$

$$\boxed{P(A) = 0.03}$$

By Baye's Theorem

$$P(A|D) = \frac{P(A) P(D|A)}{P(A)} = \frac{0.2 \times 0.06}{0.03} = 0.38$$

$$p(B|D) = \frac{P(B) p(D|B)}{P(A)} = \frac{0.3 \times 0.03}{0.031} \approx 0.290$$

$$p(C|D) = \frac{P(C) p(D|C)}{P(A)} = \frac{0.5 \times 0.02}{0.031} = 0.32$$

Q. Companies B_1, B_2, B_3 produce 30%, 45%, 25% of the cars respectively. It is known that 2%, 3%, 2% of the cars purchased from B_1, B_2, B_3 are defective.

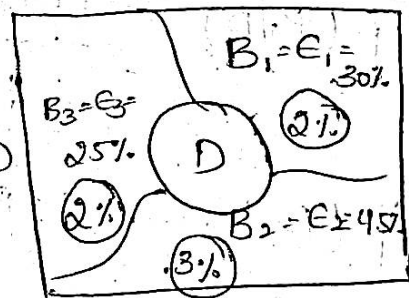
→ What is the probability that a car purchased is defective?

→ If a car purchased is found to be defective, what is the probability that this car purchased by the company B_1 ?

→ If a car purchased is found to be defective, what is the probability that this car purchased by the company B_3 ?

We know that, Bayes' Theorem

$$P(E_i|A) = \frac{P(E_i) P(A|E_i)}{\sum_{i=1}^n P(E_i) P(A|E_i)}$$



$$P(E_i|D) = \frac{P(E_i) p(D|E_i)}{\sum_{i=1}^3 P(E_i) p(D|E_i)} \quad [i=1, 2, 3]$$

$$= \frac{P(E_1) p(D|E_1)}{P(E_1) p(D|E_1) + P(E_2) p(D|E_2) + P(E_3) p(D|E_3)} \rightarrow 1$$

$$P(E_1) p(D|E_1) + P(E_2) p(D|E_2) + P(E_3) p(D|E_3)$$

$$= \frac{P(\epsilon_1) + P(D|\epsilon_1)}{P(A)}$$

$$P(\epsilon_1) = 30\% = \frac{30}{100} = 0.3$$

$$P(\epsilon_2) = 45\% = \frac{45}{100} = 0.45$$

$$P(\epsilon_3) = 25\% = \frac{25}{100} = 0.25$$

$$P(D|\epsilon_1) = 2\% = \frac{2}{100} = 0.02$$

$$P(D|\epsilon_2) = 3\% = \frac{3}{100} = 0.03$$

$$P(D|\epsilon_3) = 2\% = \frac{2}{100} = 0.02$$

$$P(A) = (0.3 \times 0.02) + (0.45 \times 0.03) + (0.25 \times 0.02)$$

$$P(A) = 0.0245$$

By Bayes' Theorem

$$P(B_1|D) = \frac{P(B_1)P(D|B_1)}{P(A)} = \frac{0.3 \times 0.02}{0.0245} = 0.244$$

$$P(B_3|D) = \frac{P(B_3)P(D|B_3)}{P(A)} = \frac{0.25 \times 0.02}{0.0245} = 0.204$$

3. A businessman goes to hotels X, Y, Z 20%, 50%, 30% of the time respectively. It is known that 5%, 4%, 8% of the rooms in X, Y, Z hotel have faulty plumbing. What is the probability that businessman that room having faulty plumbing is assigned to hotel room?

4. In a Factory machine A produces 40% of the output and machine B produces 60% of the output. On the average ~~sent~~ 9 items in 1000 produced by A are defective and one item in 250 produced by B is defective. An item drawn at random from a day's output is defective. What is the probability that produced by A (or B) 0.4

5. Of the 3 men the chances that a politician next a businessman $\left(\begin{array}{l} \frac{9}{100} = A \\ \frac{1}{250} = B \end{array} \right.$ or an academic will be appointed as a VC of a university are 0.5, 0.3, 0.2 respectively. Probability that research is promoted by these persons if they are appointed as VC are 0.3, 0.7, 0.8 respectively.

→ What is the probability that research is promoted

→ If research is promoted, what is the probability that VC is academic. (C₃)?

3. We know that, Baye's Theorem

$$P(E_i|A) = \frac{P(E_i) P(A|E_i)}{\sum_{i=1}^3 P(E_i) P(A|E_i)}$$

$$P(E_i|D) = \frac{P(E_i) P(D|E_i)}{\sum_{i=1}^3 P(E_i) P(D|E_i)}$$

$$= \frac{P(E_i) P(D|E_i)}{P(A)}$$

$$P(E_1)P(D|E_1) + P(E_2)P(D|E_2) + P(E_3)P(D|E_3)$$

$$= \frac{P(E_i) P(D|E_i)}{P(A)}$$

$$P(E_1) = 20\% = \frac{20}{100} = 0.2$$

$$P(E_2) = 50\% = \frac{50}{100} = 0.5$$

$$P(E_3) = 30\% = \frac{30}{100} = 0.3$$

$$P(D|E_1) = 5\% = 0.05$$

$$P(D|E_2) = 4\% = 0.04$$

$$P(D|E_3) = 8\% = 0.08$$

$$P(A) = 0.2 \times 0.05 + 0.5 \times 0.04 + 0.3 \times 0.08$$

$$P(A) = 0.054$$

By Baye's Theorem

$$P(Z|D) = P(E_3|D) = \frac{P(E_3) P(D|E_3)}{P(A)} = \frac{(0.3)(0.08)}{0.054} = 0.44$$

4) We know that, Baye's Theorem

$$P(E_i|A) = \frac{P(E_i) P(A|E_i)}{\sum_{i=1}^n P(E_i) P(A|E_i)}$$

$$P(E_i|D) = \frac{P(E_i) P(D|E_i)}{\sum_{i=1}^2 P(E_i) P(D|E_i)}$$

$$= \frac{P(E_i) P(D|E_i)}{P(E_1) P(D|E_1) + P(E_2) P(D|E_2)}$$
$$= \frac{P(E_i) P(D|E_i)}{P(A)}$$

$$P(E_1) = \frac{40}{100} = 0.4 \quad P(D|E_1) = 0.009 = \frac{9}{1000}$$

$$P(E_2) = \frac{60}{100} = 0.6 \quad P(D|E_2) = \frac{1}{250} = 0.004$$

$$P(D|E_1) =$$

$$P(A) = 0.4 \times 0.009 + 0.6 \times 0.004$$

$$P(A) = 0.006$$

By Baye's Theorem

$$P(A|D) = \frac{P(E_1) P(D|E_1)}{P(A)} = \frac{(0.4)(0.009)}{0.006} = 0.6$$

$$P(B|D) = \frac{P(E_2) P(D|E_2)}{P(A)} = \frac{(0.6)(0.004)}{0.006} = 0.4$$

5) We know that, Baye's Theorem

$$P(E_i|A) = \frac{P(E_i) P(A|E_i)}{\sum_{i=1}^n P(E_i) P(A|E_i)}$$

$$P(E_i|D) = \frac{P(E_i) P(D|E_i)}{\sum_{i=1}^3 P(E_i) P(D|E_i)}$$

$$= \frac{P(E_1)P(D|E_1)}{P(E_1)P(D|E_1) + P(E_2)P(D|E_2) + P(E_3)P(D|E_3)}$$

but

$$P(E_1) = 0.5$$

$$P(D|E_1) = 0.3$$

$$P(E_2) = 0.3$$

$$P(D|E_2) = 0.7$$

$$P(E_3) = 0.2$$

$$P(D|E_3) = 0.8$$

$$P(A) = 0.5 \times 0.3 + 0.3 \times 0.7 + 0.2 \times 0.8$$

$$P(A) = 0.52$$

By Baye's Theorem,

$$P(E_3/D) = \frac{P(E_3)P(D|E_3)}{P(A)} = \frac{(0.2)(0.8)}{0.52} = \underline{\underline{0.30}}$$

52 Cards

Spade (♠)

A

2

3

4

5

6

7

8

9

10

J → Black cards
Q → Black cards

K

Hearts (♥)

A

2

3

4

5

6

7

8

9

10

J → Red cards
Q → Red cards

K

Clavor (♣)

A

2

3

4

5

6

7

8

9

10

J → Black cards
Q → Black cards

K

Diamond (♦)

A

2

3

4

5

6

7

8

9

10

J → Red cards
Q → Red cards

K

1. One card is drawn from a pack of 52 cards.
What is the probability that it is either a king
or a queen

let $A = \text{king}$

$B = \text{queen}$ $n(S) = 52C_1$

$$\text{Now } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A) = \frac{n(A)}{n(S)} \quad P(B) = \frac{n(B)}{n(S)}$$

$$= \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)} - 0 \quad (A \& B \text{ are disjoint})$$

Q=103] Random Variables

Random variable: A random variable is a rule that assigns a numerical value to each possible outcome of an experiment.

i.e., outcomes related to the real line.

Example: Suppose two coins are tossed simultaneously then the sample space

$$S = \{HH, HT, TH, TT\}$$

Let X denote the number of heads then

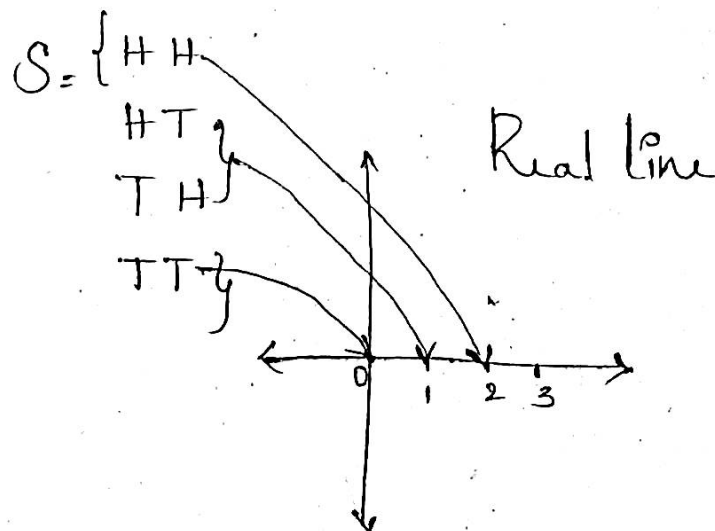
If $X=0$, then the outcomes are $\{TT\}$ & $P(X=0) = \frac{1}{4}$

If $X=1$, then the outcomes are $\{HT, TH\}$ & $P(X=1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

If $X=2$, then the outcomes are $\{HH\}$ & $P(X=2) = \frac{1}{4}$

The probability distribution of this random variable X is given by

$X = x$	0	1	2
$P(X=x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$



→ Types of Random variables

* Discrete Random variables (Finite values)

* Continuous Random variables (Infinite values)

Probability mass function (Discrete nature):

→ $\sum_{i=1}^n P_i = 1$ (or) $\sum p_i = 1$ (or) $\sum P(X=x_i) = 1$
[to find constant]

→ Cumulative Distribution function (or)
Distribution function

$$F(x) = P(X \leq x) = \sum_{t \leq x} P(t)$$

→ mean

$$E(x) = \sum x_i p_i$$

→ Variance

$$\text{Var}(x) = E(x^2) - (E(x))^2$$

$$E(x^2) = \sum x_i^2 P_i - (\mu)^2$$

$$\text{Standard deviation} = \sqrt{\text{Variance}}$$

Note:

$$\rightarrow P(X \geq a) = 1 - P(X < a)$$

$$\rightarrow P(X > a) = 1 - P(X \leq a)$$

1. A Random variable x has the following distribution function

x	0	1	2	3	4	5	6	7
$P(X=x)$	0	k	$2k$	$3k$	$3k^2$	$2k^2$	$2k^2$	$7k^2+k$
	0	$1/10$	$2/10$	$4/10$	$3/10$	$1/100$	$1/100$	$17/100$

(i) Find k (ii) Find $P(X < 6)$, $P(X \geq 6)$, $P(0 < X < 5)$, $P(0 \leq X \leq 4)$ (iii) If $P(X \leq k) = \frac{1}{2}$ find the minimum value of k and (iv) Determine the distribution function of X (v) mean (vi) variance

(i) To Find k

We know that

$$\sum P_i = 1$$

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$10k^2 + 9k = 1$$

$$10k^2 + 9k - 1 = 0$$

$$10k^2 + 10k + k - 1 = 0$$

$$10(k^2 + k)$$

$$a = 10, b = 9, c = -1$$

$$k = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-9 \pm \sqrt{81 + 40}}{2 \times 10} = \frac{-9 \pm \sqrt{121}}{20} = \frac{-9 \pm 11}{20}$$

$$k = \frac{2}{20}, \frac{-20}{20}$$

$$k = \frac{1}{10}, -1$$

$$\boxed{k = \frac{1}{10}}$$

$$(ii) P(X < 6) = P(X = 0, 1, 2, 3, 4, 5)$$

$$= 0 + k + 2k + 2k + 3k + k^2$$

$$= 8k + k^2$$

$$= 8\left(\frac{1}{10}\right) + \left(\frac{1}{10}\right)^2$$

$$= \frac{8}{10} + \frac{1}{100}$$

$$= \frac{80+1}{100} = \frac{81}{100}$$

$$P(X < 6) = \frac{81}{100}$$

$$\begin{aligned} P(X \geq 6) &= P(X=6, 7) \\ &= 2k^2 + 7k^2 + k \\ &= 9k^2 + k \\ &= 9\left(\frac{1}{10}\right)^2 + \frac{1}{10} \\ &= 9\left(\frac{1}{100}\right) + \frac{1}{10} \\ &= \frac{9}{100} + \frac{1}{10} \end{aligned}$$

$$P(X \geq 6) = \frac{19}{100}$$

$$\begin{aligned} P(0 < X < 5) &= P(X=1, 2, 3, 4) \\ &= k + 2k + 2k + 3k \\ &= 8k = \frac{8}{10} \end{aligned}$$

$$P(0 \leq X \leq 5) = \frac{8}{10}$$

$$\begin{aligned} P(0 \leq X \leq 4) &= P(X=0, 1, 2, 3, 4) \\ &= 0 + k + 2k + 2k + 3k \\ &= 8k \end{aligned}$$

$$P(0 \leq X \leq 4) = \frac{8}{10}$$

(iv) Cumulative distribution function of X

X	0	1	2	3	4	5	6	7
P(X)	0	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{5}{10}$	$\frac{8}{10}$	$\frac{81}{100}$	$\frac{83}{100}$	$\frac{100}{100} = 1$

Sum

(iii) If $p(x \leq k) > \frac{1}{2}$ find the minimum value of k

put $k=0$, $p(x \leq 0) = 0 > \frac{1}{2} \times$

put $k=1$, $p(x \leq 1) = p(x=0) + p(x=1)$
 $= 0 + k = \frac{1}{10} = 0.1 > \frac{1}{2} \times$

put $k=2$, $p(x \leq 2) = p(x=0) + p(x=1) + p(x=2)$
 $= 0 + k + 2k = 3/k$
 $= \frac{3}{10} = 0.3 > 0.5 \times$

put $k=3$, $p(x \leq 3) = p(x=0) + p(x=1) + p(x=2) + p(x=3)$
 $= 0 + k + 2k + 3k = 6k = \frac{6}{10} > \frac{1}{2} \times$

put $k=4$, $p(x \leq 4) = p(x=0) + p(x=1) + p(x=2) + p(x=3) + p(x=4)$
 $= 0 + k + 2k + 3k + 4k = 10k$
 $= \frac{8}{10} = 0.8 > 0.5 \checkmark$

$\boxed{k=4}$

(iv) mean

$E(x) = \sum x_i p_i$
 $= (0 \times 0) + (1 \times k) + (2 \times 2k) + (3 \times 3k) + (4 \times 4k) + (5 \times 5k)$
 $+ (6 \times 6k) + (7 \times 7k)$

$= 0 + k + 4k + 9k + 16k + 25k + 36k + 49k$

$= 110k$

$= 110 \left(\frac{1}{10}\right) = 11$

$= \frac{110}{10} = 11$

$= \frac{110}{10} = 11$

$$(vi) \text{ Variance } E(x) = E(x^2) - (E(x))^2$$

(Or)

$$\text{Var}(x) = \sum x_i^2 P_i - (\mu)^2$$

$$= (0^2 \times 0) + (1^2 \times k) + (2^2 \times 2k) + 3^2 \times 2k + 4^2 \times 3k + 5^2 \times k^2 + 6^2 \times 2k^2 + 7^2 \times (7k^2 + k) - (3.66)^2$$

$$= 0 + k + 8k + 18k + 48k + 25k^2 + 42k^2 + 343k^2 + 49k - (13.395)$$

$$= 440k^2 + 124k - 13.395$$

$$= 4 \cdot 4 + 12 \cdot 4 - 13.395$$

$$= \underline{\underline{3.40}}$$

Q. Find the probability distribution function of a random variate X is

$$X = x \quad 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6$$

$$P(X=x) \quad k \quad 3k \quad 5k \quad 7k \quad 9k \quad 11k \quad 13k$$

(i) Find k (ii) Find $P(x < 4)$, $P(3 < x \leq 6)$, $P(x \geq 5)$

(iii) what will be the minimum value of k , so that $P(x \leq 2) > 0.3$?

(i) To Find k

We know that

$$\sum P_i = 1$$

$$k + 3k + 5k + 7k + 9k + 11k + 13k = 1$$

$$49k = 1$$

$$k = \frac{1}{49}$$

$$(ii) p(x < 4) = p(x = 0, 1, 2, 3)$$

$$= k + 3k + 5k + 7k$$

$$= 16k = 16\left(\frac{1}{49}\right) = 0.326$$

$$p(3 < x \leq 6) = p(x = 4, 5, 6)$$

$$= 9k + 11k + 13k$$

$$= 33k = \frac{33}{49}$$

$$p(x \geq 3) = p(0, 1, 2)$$

$$= k + 3k +$$

$$p(x \geq 5) = p(x = 5, 6)$$

$$= 11k + 13k$$

$$= \frac{24}{49}$$

$$(iii) p(x \leq 2) > 0.3$$

$$p(0, 1, 2) > 0.3$$

$$k + 3k + 5k > 0.3$$

$$9k > 0.3$$

$$9k > \frac{3}{10}$$

$$k > \frac{3}{10 \times 9}$$

$$\underline{\underline{k > \frac{1}{30}}}$$

Expected value of a Discrete Random variable

$$\rightarrow E(X) = \sum x_i p_i$$

$$\rightarrow E(ax + b) = aE(x) + b$$

$$\rightarrow E(x^2) = \sum x_i^2 p_i$$

$$\rightarrow E(ax + by) = aE(x) + bE(y)$$

$$\rightarrow E(c) = c$$

1. Find the probability

1. A Variate x has the following probability distribution.

$x = x$ -3 6 9

$p(x=x)$ $\frac{1}{6}$ $\frac{1}{2}$ $\frac{1}{3}$

(i) mean (ii) variance

$\downarrow$
 $E(x) = ?$

$\downarrow$
 $E(x^2) - (E(x))^2$

Find (i) $E(x)$

(ii) $E(x^2)$

(iii) $E[(2x+1)^2]$

$\downarrow$

$$E(4x^2 + 4x + 1)$$

$$= 4E(x^2) +$$

$$4E(x) + E(1)$$

Continuous Random Variable

Define Continuous Random Variable :

When a random variable 'X' takes every value in an interval

Example : temperature, height

probability density function :

$$\rightarrow f(x) \geq 0$$

$$\rightarrow \int_{-\infty}^{\infty} f(x) dx = 1 \quad [\text{To find constant value}]$$

Results :

$$\rightarrow P(a < x \leq b) = \int_a^b f(x) dx$$

$$\rightarrow P(x > a) = \int_a^{\infty} f(x) dx$$

$$\rightarrow P(x \leq a) = \int_{-\infty}^a f(x) dx$$

Mean :

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

Variance of X :

$$\text{Var}(X) = E(x^2) - [E(x)]^2$$

Probability Distribution function :

The cumulative distribution function of a Continuous Random Variable 'x' is denoted by $F(x)$ and is defined as $F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$
Where $F(x)$ = Cumulative Distribution function

$f(x)$ = probability distribution function.

Thus $F(x)$ gives the probability the value of the variable x will be $\leq x$

properties of $F(x)$:

$$\rightarrow 0 \leq F(x) \leq 1$$

$$\rightarrow F'(x) = f(x)$$

$$\rightarrow F(-\infty) = 0$$

$$\rightarrow F(\infty) = 1$$

$$\rightarrow p(a < x < b) = F(b) - F(a) \\ = \int_a^b f(x) dx$$

1. A continuous Random Variable x has the probability density function given by $f(x) = kx^2$
 $0 \leq x \leq 1$

$\rightarrow$ Find the value of k

$$\rightarrow p(x \leq 1/2)$$

$$\rightarrow p(x \geq 3/4)$$

Sol (i) TO Find the value of k :

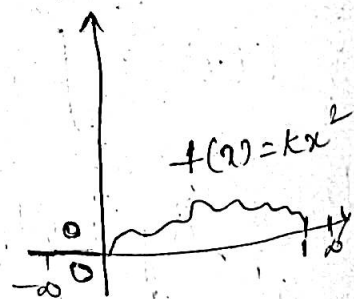
We know that

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^0 + \int_0^1 f(x) dx + \int_1^{\infty} = 1$$

$$0 + \int_0^1 kx^2 dx + 0 = 1$$

$$k \int_0^1 x^2 dx = 1$$



$$k \cdot f(x) = 1$$

$$2 \cdot k = 1$$

$$k = \frac{1}{2}$$

$$k \left[\frac{x^3}{3} \right]_0^1 = 1$$

$$k \left[\frac{1}{3} - 0 \right] = 1$$

$$\frac{k}{3} = 1$$

$$\boxed{k=3}$$

$$\therefore f(x) = kx^2, 0 \leq x \leq 1$$

$$\boxed{f(x) = 3x^2, 0 \leq x \leq 1}$$

$$(ii) p(x \leq 1/2)$$

We know that

$$p(x < a) = \int_{-\infty}^a f(x) dx$$

$$= \int_{-\infty}^{1/2} f(x) dx$$

$$= \int_{-\infty}^0 + \int_0^{1/2} f(x) dx$$

$$= 0 + \int_0^{1/2} 3x^2 dx$$

$$= 3 \left[\frac{x^3}{3} \right]_0^{1/2}$$

$$= (x^3)_0^{1/2}$$

$$= \left(\frac{1}{2} \right)^3 - 0$$

$$\boxed{p(x \leq 1/2) = \frac{1}{8}}$$

$$(iii) p(x \geq 3/4)$$

We know that

$$p(x \geq a) = \int_a^{\infty} f(x) dx$$

$$= \int_{3/4}^{\infty} f(x) dx$$

$$= \int_{3/4}^1 f(x) dx + \int_1^{\infty} f(x) dx$$

$$= \int_{3/4}^1 3x^2 dx + \int_1^{\infty} 0 dx$$

$$= 3 \left[\frac{x^3}{3} \right]_{3/4}^1$$

$$= 1^3 - \left(\frac{3}{4} \right)^3$$

$$= 1 - \frac{27}{64}$$

$$= \frac{64 - 27}{64}$$

$$= \frac{37}{64}$$

$$\boxed{p(x \geq a) = \frac{37}{64}}$$

3. A continuous Random Variable X has the probability density function given

$$f(x) = \begin{cases} kx, & 0 \leq x \leq 1 \\ k, & 1 \leq x \leq 2 \\ 3k - kx, & 2 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

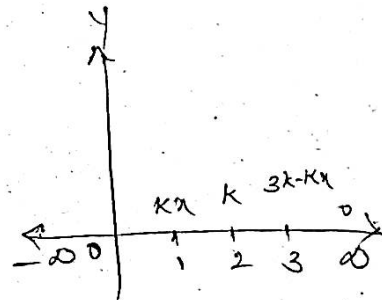
Find the value of k

Find $p(X \leq 1.5)$

Sol: To Find the value of k :

We know that

$$\int_{-\infty}^{\infty} f(x) dx = 1$$



$$\int_{-\infty}^0 + \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^3 f(x) dx + \int_3^{\infty} f(x) dx = 1$$

$$0 + \int_0^1 kx dx + \int_1^2 k dx + \int_2^3 (3k - kx) dx + \int_3^{\infty} 0 dx = 1$$

$$\int_0^1 kx dx + \int_1^2 k dx + \int_2^3 (3k - kx) dx = 1$$

$$k \left[\frac{x^2}{2} \right]_0^1 + k [x]_1^2 + 3k \left[x \right]_2^3 - k \left[\frac{x^2}{2} \right]_2^3 = 1$$

$$k \left[\frac{1}{2} \right] + k [2 - 1] + 3k (3 - 2) - \frac{k}{2} \left(\frac{9}{2} - 4 \right) = 1$$

$$\frac{k}{2} + k + 3k - \frac{k}{2} (5) = 1$$

$$\frac{k}{2} + k + 3k - \frac{5k}{2} = 1$$

$$-\frac{4k}{2} + 4k = 1$$

$$\frac{k + 2k + 6k - 5k}{2} = 1$$

2

$$\frac{9k - 5k}{2} = 1$$

$$\frac{4K}{2} = 1$$

$$4K = 2$$

$$\boxed{K = 1/2}$$

Therefore,

$$F(x) = \begin{cases} x/2, & 0 \leq x \leq 1 \\ 1/2, & 1 \leq x \leq 2 \\ \frac{3}{2} - \frac{x}{2}, & 2 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

(ii) TO Find $P(x \leq 1.5)$

We know that

$$P(x \leq a) = \int_{-\infty}^a f(x) dx$$

$$P(x \leq 1.5) = \int_{-\infty}^{1.5} f(x) dx$$

$$= \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^{1.5} f(x) dx$$

$$= 0 + \int_0^1 kx dx + \int_1^{1.5} 1/2 dx$$

$$= \int_0^1 \frac{x}{2} dx + \int_1^{1.5} \frac{1}{2} dx$$

$$= \frac{1}{2} \left[\frac{x^2}{2} \right]_0^1 + \frac{1}{2} [x]_1^{1.5}$$

$$= \frac{1}{2} \left[\frac{1}{2} \right] + \frac{1}{2} [1.5 - 1]$$

$$= \frac{1}{4} + \frac{1}{2} [0.5]$$

$$P(x \leq 1.5) = 0.5$$

Q. If x is a Continuous Random Variable with the probability density function given by

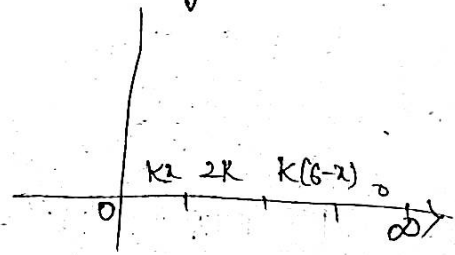
$$f(x) = \begin{cases} kx, & 0 \leq x < 2 \\ 2k, & 2 \leq x \leq 4 \\ k(6-x), & 4 \leq x \leq 6 \end{cases}$$

Find k ; mean of the density function

Sol, To find k

We know that

$$\int_{-\infty}^{\infty} f(x) dx = 1$$



$$\begin{aligned} \int_{-\infty}^0 0 dx + \int_0^2 kx dx + \int_2^4 2k dx + \int_4^6 k(6-x) dx + \int_6^{\infty} 0 dx &= 1 \\ 0 + \int_0^2 kx dx + \int_2^4 2k dx + \int_4^6 k(6-x) dx + 0 &= 1 \end{aligned}$$

$$k \left[\frac{x^2}{2} \right]_0^2 + 2k \left[x \right]_2^4 + 6k \left[x \right]_4^6 - k \left[\frac{x^2}{2} \right]_4^6 = 1$$

$$k \left[\frac{4}{2} - 0 \right] + 2k [4 - 2] + 6k [6 - 4] - k \left[\frac{36}{2} - \frac{16}{2} \right] = 1$$

$$2k + 4k + 12k - k[10] = 1$$

$$8k + 4k + 2k = 1$$

$$8k = 1$$

$$\boxed{k = \frac{1}{8}}$$

(ii) mean =

Therefore

$$F(x) = \begin{cases} \frac{x^2}{8}, & 0 \leq x \leq 2 \\ \frac{2}{8}, & 2 \leq x \leq 4 \\ -\frac{1}{8}(6-x), & 4 \leq x \leq 6 \end{cases}$$

$$(ii) \text{ mean} = E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$\int_{-\infty}^0 + \int_0^2 x f(x) dx + \int_2^4 x f(x) dx + \int_4^6 x f(x) dx + \int_6^{\infty} = 1$$

$$0 + \int_0^2 x \left(\frac{x}{8}\right) dx + \int_2^4 x \left(\frac{x}{4}\right) dx + \int_4^6 x \left(\frac{1}{8}(6-x)\right) dx + 0 = 1$$

$$\Rightarrow \frac{1}{8} \int_0^2 x^2 dx + \frac{1}{4} \int_2^4 x^2 dx + \frac{1}{8} \int_4^6 x(6-x) dx = 1$$

$$\Rightarrow \frac{1}{8} \left[\frac{x^3}{3} \right]_0^2 + \frac{1}{4} \left[\frac{x^3}{3} \right]_2^4 + \frac{1}{8} \int_4^6 6x - x^2 dx = 1$$

$$\Rightarrow \frac{1}{8} \left[\frac{2^3}{3} - 0 \right] + \frac{1}{4} \left[\frac{4^3}{3} - \frac{2^3}{3} \right] + \frac{1}{8} \left[6 \left[\frac{x^2}{2} \right]_4^6 - \left[\frac{x^3}{3} \right]_4^6 \right] = 1$$

$$\Rightarrow \frac{1}{8} \left[\frac{8}{3} - 0 \right] + \frac{1}{4} \left[\frac{64}{3} - \frac{8}{3} \right] + \frac{1}{8} \left[6 \left(\frac{36}{2} - \frac{16}{2} \right) - \left(\frac{6^3}{3} - \frac{4^3}{3} \right) \right] = 1$$

$$\frac{1}{8} \times \frac{8}{3} + \frac{1}{4} [4] + \frac{1}{8} [$$

$$\Rightarrow \underline{\underline{\frac{1}{3}}}$$

5) If the probability density function of a random variable is given by

$$f(x) = \begin{cases} k(1-x^2), & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \text{ find } k$$

$$(ii) p(0.1 < x < 0.2)$$

$$(iii) p(x > 0.5)$$

6) A continuous random variable x has the

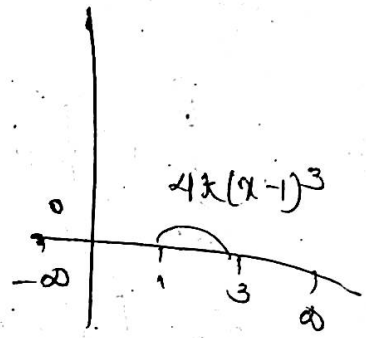
$$\text{distribution function } f(x) = \begin{cases} 0, & x \leq 1 \\ k(x-1)^4, & 1 < x < 3 \\ 1, & x \geq 3 \end{cases}$$

probability density function

To find $f(x)$

$$f(x) = \frac{d}{dx} [F(x)]$$

$$f(x) = \begin{cases} 0 & x \leq 1 \\ 4k(x-1)^3, & 1 \leq x \leq 3 \\ 0 & x > 3 \end{cases}$$



To find k

We know that $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\int_{-\infty}^1 0 dx + \int_1^3 4k(x-1)^3 dx + \int_3^{\infty} 0 dx = 1$$

$$0 + \int_1^3 4k(x-1)^3 dx + 0 = 1$$

$$4k \int_1^3 (x-1)^3 dx = 1$$

$$4k \left[\frac{(x-1)^4}{4} \right]_1^3 = 1$$

$$4k \left[\frac{(x-1)^4}{4} \right]_1^3 = 1$$

$$k [(x-1)^4]_1^3 = 1$$

$$k [(3-1)^4] = 1$$

$$k [2^4] = 1$$

$$k (16) = 1$$

$$k = \frac{1}{16}$$

$$F(x) = 4k(x-1)^3, \quad 1 \leq x \leq 3$$

$$= 4 \left(\frac{1}{16} \right) (x-1)^3$$

$$= \frac{1}{4}(x-1)^3, 1 \leq x \leq 3$$

To Find probability density function

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Now

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{-1} 0 + \int_{-1}^3 \frac{1}{4}(x-1)^3 dx + \int_3^{\infty} 0$$

$$= \frac{1}{4} \int_{-1}^3 (x-1)^3 dx$$

$$= \frac{1}{4} \left[\frac{(x-1)^{3+1}}{3+1} \right]_{-1}^3$$

$$= \frac{1}{4} \left(\frac{(x-1)^4}{4} \right)_{-1}^3$$

$$= \frac{1}{16} \left[(x-1)^4 \right]_{-1}^3$$

$$= \frac{1}{16} (2^4 - 0) = 1$$

7. Find a distribution function for the random variable x is defined by

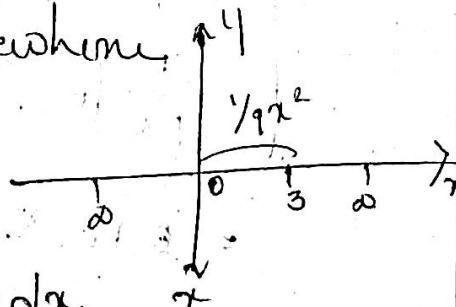
$$f(x) = \begin{cases} \frac{1}{9}x^2, & 0 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

$$\text{Given } f(x) = \begin{cases} \frac{1}{9}x^2, & 0 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

To find

1) Distribution function

$$F(x) = P(x \leq x) = \int_{-\infty}^x f(x) dx$$



$$2) p(1 \leq x \leq 2)$$

We know that

$$p(a \leq x \leq b) = f(b) - f(a)$$

$$p(a \leq x \leq b) = \int_a^b f(x) dx =$$

$$p(1 \leq x \leq 2) = f(2) - f(1)$$

$$= \frac{23}{27} - \frac{13}{27}$$

$$= \frac{8}{27} - \frac{1}{27} = \frac{7}{27}$$

8) Find the CDF of a random variable x is $F(x) = 1 - (1+x)e^{-x}$, x find the probability density function of x mean and variance of x

To find $f(x)$, we know that

$$f(x) = \frac{d}{dx} [F(x)]$$

$$f(x) = \frac{d}{dx} [1 - e^{-x} - xe^{-x}]$$

$$= \frac{d}{dx} [1] - \frac{d}{dx} (e^{-x}) - \frac{d}{dx} [xe^{-x}]$$

$$\left[\because d(uv) = u dv + v du \right]$$

$$\left[\frac{d}{dx} (e^{-ax}) = -ae^{-ax} \right]$$

$$= 0 + e^{-x} - [-xe^{-x} + e^{-x} \cdot 1]$$

$$= e^{-x} + xe^{-x} - e^{-x} = xe^{-x}, x > 0$$

$$f(x) = xe^{-x}, x > 0$$

We know that

to find mean

$$f(x) = \int_{-\infty}^{\infty} xf(x) dx$$

$$= \int_{-\infty}^0 + \int_0^{\infty} xf(x) dx = 0 + \int_0^{\infty} x(xe^{-x}) dx$$

$$= \int_0^{\infty} x^2 e^{-x} dx$$

$$\left[\because \int u dv = uv - \int u dv \right] \text{ [ILATE]}$$

$$\left[\int u dv = uv - u'v_1 + u''v_2 - u'''v_3 + \dots \right]$$

$$u = x^2, dv = e^{-x} dx$$

$$u' = 2x, v = -e^{-x}$$

$$u'' = 2, v_1 = e^{-x}$$

$$u''' = 0, v_2 = -e^{-x}$$

$$= \left[-x^2 e^{-x} - 2x e^{-x} - 2 e^{-x} \right]_0^{\infty}$$

$$= 0 - [0 - 0 - 2]$$

$$E(x) = 2$$

To find variance

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$= \int_0^{\infty} x^2 (xe^{-x}) dx = \int_0^{\infty} x^3 e^{-x} dx$$

$$E(x^2) = 6$$

$$\text{Var}(x) = E(x^2) - (E(x))^2 = 6 - (2)^2 = 6 - 4 = 2$$

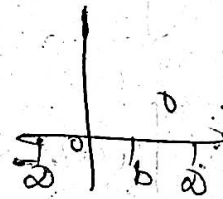
Using Bernoulli's formula

9. Find a Constant 'b' so that function

$$f(x) = \begin{cases} 3x \\ e^{3x}, & 0 \leq x \leq b \\ 0, & \text{elsewhere} \end{cases}$$

To find b, we know that

$$\int_{-\infty}^{\infty} f(x) dx = 1$$



$$\int_{-\infty}^0 0 dx + \int_0^b e^{3x} dx + \int_b^{\infty} 0 dx = 1$$

$$\frac{1}{3} \int_0^b e^{3x} dx = 1$$

$$\frac{1}{3} \left[\frac{e^{3x}}{3} \right]_0^b = 1$$

$$\frac{1}{9} (e^{3x})_0^b = 1$$

$$(e^{3x})_0^b = 9 \Rightarrow e^{3b} - e^0 = 9$$

$$e^{3b} - 1 = 9$$

$$e^{3b} = 10$$

$$3b = \ln 10$$

Taking log on both sides

$$\log(e^{3b}) = \log 10$$

$$3b = \log 10$$

$$3b = 2.303$$

$$b = \frac{2.565}{13}$$

13

$$b = 0.855$$